

Supplemental Material for “Exact Replica Mechanics and Born-Record Large Deviations in Monitored Quantum Circuits”

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This Supplemental Material collects the reference data behind Secs. II.F and IV and Appendix A of the main text: the exceptional sets E_L of the three-replica compressed operator for $d = 2$ and $d = 3$ with blocks, minimal-polynomial degrees and exact ranks (Sec. SI), the per-block certificate data at $L = 12$ (Sec. SII), the endpoint and $p_\times$ ranks (Sec. SIII), the complete list of certificate files and scripts with SHA-256 hashes and reproduction commands (Sec. SIV); two proofs deferred from the main text (Secs. SV and SVI); the Clifford record-count runs behind the main-text Sec. IV.F (Sec. SVII), and the exact-closure suite of that section (the shared `gap_utils` library with its validation battery, the conjugation-design test, the exact $\bar{Z}_3$ cells and ladder, the $\beta = 2$ identification cross-check, the marginal $q = 4$ log-correction fits, the tilted-variance profile family, and the symmetry-reduced $n = 5$ $L = 8$ block route; Sec. SVIID). Every number is exact (integer, modular with the stated prime, or algebraic with a certified isolating interval); no floating-point computation enters. Notation follows the main text: B_p is the integer-polynomial matrix of Eq. (A1), $\Gamma = H \times \mathbb{Z}_m$ with $H = (S_3 \times S_3) \rtimes \mathbb{Z}_2$, blocks are labelled (ρ, k) by an irreducible representation ρ of H (dimension d_ρ ; $\iota\pm$ is the transposition parity, Ind an induced representation of dimension 2 or 4) and a momentum k of $\mathbb{Z}_m$ ($k = \pm 1$ and $k = \pm 2$ denote the rational block that is the sum of the two conjugate complex sectors).

SI. EXCEPTIONAL SETS

At a point $p^* \in E_L$ the drop of rank B_{p^*} below the generic rank is $\sum d_\rho \nu_\rho$, where ν_ρ is the nullity of the block at p^* ; the tables list, for each p^* , the block(s) whose invariant factor vanishes, the degree of the minimal polynomial f of p^* over $\mathbb{Q}$, the nullity, and the exact rank of B_{p^*} . The numerical value of p^* is a truncation of the certified isolating interval (all intervals have width below 10^{-15}). For $d = 2$ every interior zero is simple (certificate multiplicity equal to the nullity in all 11 block factors at $L = 8$ and the one at $L = 6$); for $d = 3$ all are simple except the linear factor $3p - 2$ ($p_\times(3) = 2/3$) in the $k = \pm 1$ blocks $(\text{triv} \otimes \text{triv})^{\iota+}$ at $L = 6$, $(\text{std} \otimes \text{std})^{\iota+}$ at $L = 6, 8$ and $(\text{sgn} \otimes \text{sgn})^{\iota+}$ at $L = 8$ (file `n3_simple_zero_check.log`).

TABLE S1. E_8 for $d = 2$ (generic rank 1202): the eleven algebraic points in $(0, 1)$. For $L = 6$, $d = 2$, the single point is $p^* = (9 - \sqrt{41})/4 = 0.649219\dots$, block $\text{Ind}(\text{sgn} \otimes \text{std})$, $k = \pm 1$, nullity 2, rank 208; $E_4 = \emptyset$.

p^*	block ρ	k	$\deg f$	nullity	rank
0.255038	$\text{Ind}(\text{triv} \otimes \text{std})$	0	9	1	1198
0.270503	$(\text{std} \otimes \text{std})^{\iota+}$	0	14	1	1198
0.381759	$(\text{triv} \otimes \text{triv})^{\iota+}$	0	19	1	1201
0.404429	$\text{Ind}(\text{triv} \otimes \text{std})$	0	8	1	1198
0.510263	$(\text{std} \otimes \text{std})^{\iota-}$	± 1	6	2	1194
0.548044	$(\text{triv} \otimes \text{triv})^{\iota+}$	± 1	11	2	1200
0.557093	$\text{Ind}(\text{triv} \otimes \text{std})$	± 1	12	2	1194
0.684029	$\text{Ind}(\text{triv} \otimes \text{std})$	2	5	2	1194
0.775468	$(\text{sgn} \otimes \text{sgn})^{\iota+}$	± 1	11	2	1200
0.862494	$\text{Ind}(\text{triv} \otimes \text{std})$	± 1	12	2	1194
0.960754	$(\text{sgn} \otimes \text{sgn})^{\iota+}$	0	22	1	1201

TABLE S2. Exceptional points for $d = 3$ at $L = 4$ (generic rank 21) and $L = 6$ (generic rank 216). At $L = 6$ the rational point $p_\times(3) = 2/3$ affects seven blocks simultaneously— $(\text{triv} \otimes \text{triv})^{\iota+}$, $k = \pm 1$ (nullity 2); $\text{Ind}(\text{triv} \otimes \text{sgn})$, $k = 0$ (1) and $k = \pm 1$ (2); $(\text{std} \otimes \text{std})^{\iota+}$, $k = 0$ (1) and $k = \pm 1$ (4); $\text{Ind}(\text{triv} \otimes \text{std})$, $k = 0$ (1) and $k = \pm 1$ (2)—with total drop 40 and rank 176.

L	p^*	block ρ	k	$\deg f$	ν	drop	rank
4	0.779812	$(\text{sgn} \otimes \text{sgn})^{\iota+}$	0	5	1	1	20
6	0.393762	$\text{Ind}(\text{sgn} \otimes \text{std})$	± 1	4	2	8	208
6	0.617321	$(\text{sgn} \otimes \text{sgn})^{\iota+}$	0	14	1	1	215
6	2/3	seven blocks (caption)		1		40	176
6	0.744876	$(\text{triv} \otimes \text{triv})^{\iota+}$	0	16	1	1	215
6	0.775117	$(\text{std} \otimes \text{std})^{\iota+}$	0	13	1	4	212
6	0.826750	$(\text{std} \otimes \text{std})^{\iota+}$	± 1	2	2	8	208

SII. THE $L = 12$ CERTIFICATE, BLOCK BY BLOCK

Table S4 lists, for the momentum- π sector $k = 3$ of $L = 12$ ($m = 6$), the multiplicity $m_{\rho,3}$, the reflection-parity dimensions (m_+, m_-) , the symmetry-forced bound $2 \min(m_+, m_-)$, the numbers (c_+, c_-) of exactly verified kernel vectors on the two parity sides, the resulting upper bound $2 \min(m_+ - c_+, m_- - c_-)$ valid for all p outside a finite set, and the modular lower bound at $p = 1/2$; the two bounds coincide in every block. The sectors $k \neq 3$ have

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TABLE S3. E_8 for $d = 3$ (generic rank 1202): 55 points. At $p_\times(3) = 2/3$ seven blocks lose rank simultaneously— $(\text{sgn} \otimes \text{sgn})^{\iota+}$, $k = \pm 1$ (nullity 2); $\text{Ind}(\text{triv} \otimes \text{sgn})$, $k = 0$ (1) and $k = \pm 1$ (4); $(\text{std} \otimes \text{std})^{\iota+}$, $k = 0$ (1) and $k = \pm 1$ (6); $\text{Ind}(\text{sgn} \otimes \text{std})$, $k = 0$ (1) and $k = \pm 1$ (4)—for a total drop of 60 and rank 1142. For the two degree-46 and degree-50 factors of $(\text{std} \otimes \text{std})^{\iota+}$ the nullity is pinned by $1 \leq \nu \leq \text{mult}_f(\det)$ and the evenness of ν in $k = \pm 1$ blocks, as stated in Appendix A.3.

p^*	block ρ, k	deg f	ν	drop	rank	p^*	block ρ, k	deg f	ν	drop	rank
0.019707	$\text{Ind}(\text{sgn} \otimes \text{std}), 0$	11	1	4	1198	0.698975	$\text{Ind}(\text{sgn} \otimes \text{std}), 0$	14	1	4	1198
0.043971	$(\text{triv} \otimes \text{triv})^{\iota+}, 0$	49	1	1	1201	0.699932	$(\text{std} \otimes \text{std})^{\iota+}, \pm 1$	46	2	8	1194
0.059680	$(\text{triv} \otimes \text{triv})^{\iota+}, \pm 1$	25	2	2	1200	0.717603	$\text{Ind}(\text{triv} \otimes \text{sgn}), 0$	6	1	2	1200
0.114953	$(\text{std} \otimes \text{std})^{\iota-}, \pm 1$	10	2	8	1194	0.726117	$(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$	40	1	1	1201
0.116570	$\text{Ind}(\text{triv} \otimes \text{std}), 0$	23	1	4	1198	0.727191	$(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$	40	1	1	1201
0.157295	$(\text{std} \otimes \text{std})^{\iota+}, \pm 1$	46	2	8	1194	0.727894	$\text{Ind}(\text{triv} \otimes \text{std}), 0$	12	1	4	1198
0.280527	$\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$	26	2	8	1194	0.734664	$(\text{std} \otimes \text{std})^{\iota+}, \pm 1$	46	2	8	1194
0.327751	$(\text{std} \otimes \text{std})^{\iota+}, \pm 1$	46	2	8	1194	0.736197	$\text{Ind}(\text{sgn} \otimes \text{std}), 0$	11	1	4	1198
0.387717	$\text{Ind}(\text{triv} \otimes \text{std}), 2$	10	2	8	1194	0.739118	$\text{Ind}(\text{sgn} \otimes \text{std}), \pm 1$	28	2	8	1194
0.406403	$(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$	40	1	1	1201	0.747220	$\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$	26	2	8	1194
0.412493	$\text{Ind}(\text{triv} \otimes \text{std}), 2$	10	2	8	1194	0.750329	$(\text{std} \otimes \text{std})^{\iota-}, \pm 1$	10	2	8	1194
0.449885	$(\text{std} \otimes \text{std})^{\iota-}, 0$	7	1	4	1198	0.752473	$(\text{std} \otimes \text{std})^{\iota+}, 0$	50	1	4	1198
0.451387	$(\text{std} \otimes \text{std})^{\iota+}, 0$	7	1	4	1198	0.778267	$(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$	40	1	1	1201
0.482738	$(\text{std} \otimes \text{std})^{\iota-}, 0$	4	1	4	1198	0.803763	$(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$	40	1	1	1201
0.488163	$(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$	3	1	1	1201	0.818251	$\text{Ind}(\text{sgn} \otimes \text{std}), 0$	14	1	4	1198
0.499209	$\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$	26	2	8	1194	0.819702	$(\text{std} \otimes \text{std})^{\iota+}, \pm 1$	46	2	8	1194
0.562734	$\text{Ind}(\text{triv} \otimes \text{std}), 0$	12	1	4	1198	0.826750	$(\text{std} \otimes \text{std})^{\iota+}, \pm 1$	2	2	8	1194
0.604733	$\text{Ind}(\text{triv} \otimes \text{std}), 0$	23	1	4	1198	0.838695	$(\text{std} \otimes \text{std})^{\iota+}, 0$	7	1	4	1198
0.611851	$\text{Ind}(\text{sgn} \otimes \text{std}), \pm 1$	28	2	8	1194	0.839191	$(\text{std} \otimes \text{std})^{\iota+}, 0$	50	1	4	1198
0.616454	$(\text{triv} \otimes \text{triv})^{\iota+}, 2$	14	2	2	1200	0.851727	$\text{Ind}(\text{sgn} \otimes \text{std}), \pm 1$	28	2	8	1194
0.628181	$(\text{triv} \otimes \text{triv})^{\iota+}, \pm 1$	25	2	2	1200	0.882507	$(\text{std} \otimes \text{std})^{\iota+}, \pm 1$	46	2	8	1194
0.630557	$(\text{std} \otimes \text{std})^{\iota+}, 0$	50	1	4	1198	0.893682	$(\text{std} \otimes \text{std})^{\iota+}, 0$	50	1	4	1198
0.630590	$(\text{triv} \otimes \text{triv})^{\iota+}, 0$	49	1	1	1201	0.895616	$(\text{std} \otimes \text{std})^{\iota+}, 0$	7	1	4	1198
0.651297	$(\text{std} \otimes \text{std})^{\iota+}, 0$	7	1	4	1198	0.942564	$\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$	26	2	8	1194
0.658235	$(\text{std} \otimes \text{std})^{\iota+}, 0$	50	1	4	1198	0.951754	$(\text{triv} \otimes \text{triv})^{\iota+}, 0$	49	1	1	1201
0.664027	$(\text{std} \otimes \text{std})^{\iota+}, \pm 1$	46	2	8	1194	0.951943	$(\text{triv} \otimes \text{triv})^{\iota+}, 0$	3	1	1	1201
0.694261	$(\text{sgn} \otimes \text{sgn})^{\iota+}, \pm 1$	25	2	2	1200	0.985582	$\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$	26	2	8	1194

full multiplicity rank, 38906 in total, so the generic rank is $38906 + 7104 = 46010 = r_3(12) - 70$. The block data are identical for $d = 2$ and $d = 3$ (files `n3_L12_final.json`, `n3_L12_final_d3.json`).

TABLE S4. $k = 3$ sector at $L = 12$. Weighted totals: forced 7174, upper bound 7104, lower bound 7104.

ρ	d_ρ	$m_{\rho,3}$	(m_+, m_-)	forced	(c_+, c_-)	upper	lower
$(\text{triv} \otimes \text{triv})^{\iota+}$	1	162	(85,77)	154	(0,0)	154	154
$(\text{triv} \otimes \text{triv})^{\iota-}$	1	55	(19,36)	38	(0,3)	38	38
$(\text{sgn} \otimes \text{sgn})^{\iota+}$	1	170	(99,71)	142	(4,0)	142	142
$(\text{sgn} \otimes \text{sgn})^{\iota-}$	1	61	(30,31)	60	(1,0)	58	58
$\text{Ind}(\text{triv} \otimes \text{sgn})$	2	237	(132,105)	210	(4,1)	208	208
$(\text{std} \otimes \text{std})^{\iota+}$	4	545	(304,241)	482	(5,0)	482	482
$(\text{std} \otimes \text{std})^{\iota-}$	4	328	(164,164)	328	(1,4)	320	320
$\text{Ind}(\text{triv} \otimes \text{std})$	4	411	(210,201)	402	(0,3)	396	396
$\text{Ind}(\text{sgn} \otimes \text{std})$	4	423	(234,189)	378	(7,1)	376	376

Table S5 gives the modular ranks of every block (ρ, k) at $L = 12$ for three values of p : a generic point ($p = 1/2$, $d = 2$ and $d = 3$ give identical tables), the endpoint $p = 0$ ($d = 2$; identical for the second prime 1048573), and $p_\times(3) = 2/3$ at $d = 3$ (identical for the second prime 1048573). Blocks with k and $-k$ have equal rank; only $k = 0, 1, 2, 3$ are listed. Modular ranks are lower bounds for the rank over $\mathbb{Q}$; at $p = 1/2$ they meet the upper bound of Table S4.

III. ENDPOINT AND $p_\times$ RANKS

SIV. FILES, HASHES AND REPRODUCTION

All certificate files are plain JSON or text. Table S8 lists them (63 files) with their full SHA-256 hashes; the same list, in `sha256sum` format, is the ledger `certificate_sha256.txt`, so that `sha256sum`

TABLE S5. Block ranks at $L = 12$: multiplicity $m_{\rho,k}$ and rank at $p = 1/2$ (generic), $p = 0$, and $p = p_{\times}(3) = 2/3$ ($d = 3$). Weighted totals $\sum d_{\rho}$ rank over all k : 46010, 45822, 44898.

block ρ	d_{ρ}	$k=0$	$m_{\rho,k}$			rank, $p = \frac{1}{2}$				rank, $p = 0$				rank, $p = \frac{2}{3}, d = 3$			
			1	2	3	0	1	2	3	0	1	2	3	0	1	2	3
$(\text{triv} \otimes \text{triv})^{\iota+}$	1	190	160	183	162	190	160	183	154	190	160	183	154	187	158	180	154
$(\text{triv} \otimes \text{triv})^{\iota-}$	1	64	55	63	55	64	55	63	38	62	54	61	38	58	51	57	38
$(\text{sgn} \otimes \text{sgn})^{\iota+}$	1	182	166	177	170	182	166	177	142	178	163	172	142	171	158	165	142
$(\text{sgn} \otimes \text{sgn})^{\iota-}$	1	58	60	58	61	58	60	58	58	58	60	58	58	58	60	58	58
$\text{Ind}(\text{triv} \otimes \text{sgn})$	2	222	233	218	237	222	233	218	208	222	233	218	208	209	223	205	206
$(\text{std} \otimes \text{std})^{\iota+}$	4	560	545	560	545	560	545	560	482	556	542	555	482	533	525	532	480
$(\text{std} \otimes \text{std})^{\iota-}$	4	322	328	322	328	322	328	322	320	322	328	322	320	322	328	322	320
$\text{Ind}(\text{triv} \otimes \text{std})$	4	426	411	426	411	426	411	426	396	426	411	426	396	423	409	423	396
$\text{Ind}(\text{sgn} \otimes \text{std})$	4	414	423	414	423	414	423	414	376	410	420	409	376	395	409	395	374

TABLE S6. Exact ranks of B_p at the endpoints and at $p_{\times}(d) = (d+1)/[3(d-1)]$. Generic ranks are certified for $d = 2, \dots, 7$ at $L \leq 10$ and $d = 2, 3$ at $L = 12$; $p = 0$ ranks are exact over $\mathbb{Z}$ for $L \leq 8$ ($d = 2, 3$) and modular over two primes at $L = 10, 12$; $p_{\times}$ ranks are modular over two primes for $d = 3, \dots, 7$ ($L \leq 10$) and for $d = 3$ at $L = 12$ (primes 1048261 and 1048573, different seeds, identical block ranks). At $p = 1$ the rank is 1 for every L and d (Theorem 5(iii)).

L	$6^{L/2}$	$r_3(L)$	generic	$p = 0$	$p_{\times}$	$ E_L , d = 2/3$
4	36	21	21	21	21	0 / 1
6	216	216	216	198	176	1 / 6
8	1296	1202	1202	1184	1142	11 / 55
10	7776	7776	7776	7663	7276	? / ?
12	46656	46080	46010	45822	44898	? / ?

-c `certificate_sha256.txt` verifies the deposit. The later deposit rounds extend the ledger without altering any entry above: `certificate_sha256_v5.txt` (the $n = 4$ colour-resolved scans, the first $n = 5$ diagnostics, and the Haar sequential-Monte-Carlo results quoted in the main text), `certificate_sha256_v6.txt` (the $n = 5$ two-size test, the $n = 4$ $L = 10$ third crossing, and the no-freezing-theorem verification), `certificate_sha256_v8.txt` (the Clifford record-count round: the manuscript and supplement of that round together with the record-count suite whose runs are tabulated in Sec. SVII; the parallel-line ledger of that suite is `certificate_sha256_v3_scgf.txt`), and `certificate_sha256_v9.txt` (the exact-closure round: the suite of Sec. SVIID together with the `gap_utils` copies distributed alongside the suites that depend on them), each self-verifying in the same way. Table S9 lists the scripts (Python 3, dependencies `numpy` and `python-flint`; `sympy` for the $d = 2$ number-field ranks). The reproduction sequence is given in Table S7.

The $L = 12$ lower-bound runs take about 15 minutes each on one core with less than 1.5 GB of memory; all other steps take seconds to a few minutes.

SV. PROOF OF THE FREEZING THEOREM (THEOREM 8 OF THE MAIN TEXT)

The main text states the theorem in Sec. IV and defers the proof here. The statement is reproduced with equations renumbered. Let $a = \log 2$ and $s_c = \sqrt{2a} = 1.17741$. On one probability space let $\{E_{T,\omega}(Y) : T \geq 1, Y \in \{0, 1\}^T\}$ be independent Gaussian variables with $E_{T,\omega}(Y) \sim \mathcal{N}(0, T)$, and for real s put

$$\mathcal{Z}_{s,T,\omega} = \sum_{Y \in \{0,1\}^T} e^{-sE_{T,\omega}(Y)}, \quad P_{T,\omega}(Y) = \frac{e^{-E_{T,\omega}(Y)}}{\mathcal{Z}_{1,T,\omega}}. \quad (\text{S1})$$

(i) (Born realizability.) For every T and ω , $P_{T,\omega}$ is the record law of the sequential computational-basis measurement of the T qubits of the pure state $|\Psi_{T,\omega}\rangle = \sum_Y \sqrt{P_{T,\omega}(Y)} |Y\rangle$, and the predictors of Definition 1 of the main text are $r_{T,\omega,t}(y_t | y_{<t}) = P_{T,\omega}(y_{\leq t})/P_{T,\omega}(y_{<t})$, so that $\mathcal{A}_T = -\log P_{T,\omega}(Y)$ and $\phi_{T,\omega}(k) = T^{-1} \log \sum_Y P_{T,\omega}(Y)^{1-k}$ are exactly the sample log-MGFs of that definition.

(ii) (Extremes.) As $T \rightarrow \infty$, almost surely, $\max_Y (-E_{T,\omega}(Y))/T \rightarrow s_c$ and $\min_Y (-E_{T,\omega}(Y))/T \rightarrow -s_c$.

(iii) (Quenched free energy.) Almost surely, simultaneously for all real s ,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \log \mathcal{Z}_{s,T,\omega} = \varphi(s) = \begin{cases} a + s^2/2, & |s| \leq s_c, \\ |s| s_c, & |s| \geq s_c, \end{cases} \quad (\text{S2})$$

and the convergence is locally uniform in s ; φ is convex and C^1 on $\mathbb{R}$, φ'' jumps from 1 to 0 at $\pm s_c$, and $\varphi(1) = a + \frac{1}{2}$ because $1 < s_c$.

(iv) (Record exponents.) Almost surely, for every real

TABLE S7. Reproduction commands (Python 3; NB = 2, 3, 4 is the number of bonds, $L = 2\text{NB}$).

command	purpose
python3 n3_mult_certificates.py NB	$d = 2$ interval certificates (Table II of the main text)
python3 n3_mult_certificates_d.py NB 3	$d = 3$ interval certificates
python3 n3_exceptional_exact.py NB	$d = 2$ exact ranks at the exceptional points (Table S1)
python3 n3_exceptional_exact_d.py NB 3	$d = 3$ exact ranks at the exceptional points (Tables S2, S3)
python3 n3_simple_zero_check.py	order of vanishing of the certificate factors
python3 n3_p0_exact_rank.py	$p = 0$ ranks over $\mathbb{Z}$, $L \leq 8$
python3 n3_closed_form_proof.py	closed form of the Burnside deficit, Eq. (rank3)
python3 n3_sharpness_L12.py 6 [--d D --p P --q Q --seed S]	modular lower bounds, all blocks (Table S5)
python3 n3_L12_structure_check.py	antisymmetry of the $k = m/2$ sector
python3 n3_L12_kernel.py 6	constant kernel vectors ($k = 3$)
python3 n3_L12_kernel_verify.py 6	exact verification and all- p upper bound
python3 n3_L12_moving_kernel.py 6 --block B --side - [--d 3]	polynomial kernel vectors
python3 n3_L12_final.py [--d 3]	two-sided certificate (Table S4)
python3 n3_kernel_recheck.py [--d 3]	independent re-verification of all kernel vectors

k and every $q \geq 0$,

$$\psi(k) = \lim_{T \rightarrow \infty} \phi_{T,\omega}(k) = \varphi(1-k) - (1-k)\varphi(1), \quad (\text{S3})$$

$$\begin{aligned} \tau(q) &= -\psi(1-q) = q\varphi(1) - \varphi(q) \\ &= \begin{cases} q(a + \frac{1}{2}) - a - \frac{q^2}{2}, & 0 \leq q \leq q_c, \\ q(a + \frac{1}{2} - s_c), & q \geq q_c, \end{cases} \end{aligned} \quad (\text{S4})$$

with $q_c = s_c$. Thus τ is C^1 on $[0, \infty)$ with a linear branch of slope $a + \frac{1}{2} - s_c = 0.01574$ for $q \geq q_c$, τ'' jumps from -1 to 0 at q_c , and ψ is convex, finite and C^1 on $\mathbb{R}$ but not analytic at $k_c = 1 - s_c = -0.17741$ (and, on the unphysical side $q < 0$, at $1 + s_c = 2.17741$). Explicitly $\psi(k) = \frac{1}{2}k^2 + (a - \frac{1}{2})k$ for $k_c \leq k \leq 1 + s_c$ and ψ is affine outside this interval.

- (v) (Large deviations.) ψ is essentially smooth, so the main-text Theorem 7 applies to the sequence $(P_{T,\omega})_T$: the surprisal rate $h = \mathcal{A}_T/T$ satisfies the full LDP with the good rate function $I(h) = \frac{1}{2}(h - h^*)^2$ for $h \in [h_{\min}, h_{\max}]$ and $I(h) = +\infty$ otherwise, where $h^* = \psi'(0) = a - \frac{1}{2} = 0.19315$ is the typical surprisal rate, $h_{\min} = \psi'(k_c) = a + \frac{1}{2} - s_c = 0.01574$ and $h_{\max} = \psi'(1 + s_c) = a + \frac{1}{2} + s_c = 2.37056$. The singularity spectrum is $f(\alpha) = \alpha - I(\alpha)$ on $[h_{\min}, h_{\max}]$, with $f(h_{\min}) = 0$: the most probable records, $P_{T,\omega}(Y) = e^{-Th_{\min} + o(T)}$, are subexponentially many, and the tilted ensembles $P_{q,\omega}$ with $q > q_c$ all concentrate on them.

Proof. (i) The state is normalized because $\sum_Y P_{T,\omega}(Y) = 1$. Measuring qubit 1 in the computational basis gives y_1 with probability $\sum_{y_{>1}} P_{T,\omega}(y_1 y_{>1}) = P_{T,\omega}(y_1)$ and leaves the normalized state $\sum_{y_{>1}} \sqrt{P_{T,\omega}(y_1 y_{>1})/P_{T,\omega}(y_1)} |y_1 y_{>1}\rangle$; inductively the Born predictor at step t is $P_{T,\omega}(y_{\leq t})/P_{T,\omega}(y_{< t})$ and the product of the predictors telescopes to $P_{T,\omega}(Y)$. The

process is a monitored quantum process in the sense of Sec. IV of the main text with the post-measurement state as X_{t-1} and no unitary steps; it is nonlocal in the record index and its Hilbert space has dimension 2^T .

- (ii) Write $g_Y = -E_{T,\omega}(Y)/\sqrt{T}$, i.i.d. standard Gaussians, $M_T = \max_Y g_Y$, and $\bar{\Phi}(x) = \Pr(g > x)$. Upper bound: for $\varepsilon > 0$, by the union bound and $\bar{\Phi}(x) \leq e^{-x^2/2}$,

$$\Pr(M_T > \sqrt{T}(s_c + \varepsilon)) \leq 2^T e^{-T(s_c + \varepsilon)^2/2} = e^{-T(\varepsilon s_c + \varepsilon^2/2)},$$

using $2^T = e^{Ts_c^2/2}$; the right side is summable in T . Lower bound: by independence and $1 - x \leq e^{-x}$,

$$\begin{aligned} \Pr(M_T < \sqrt{T}(s_c - \varepsilon)) &= (1 - \bar{\Phi}(\sqrt{T}(s_c - \varepsilon)))^{2^T} \\ &\leq \exp(-2^T \bar{\Phi}(\sqrt{T}(s_c - \varepsilon))), \end{aligned}$$

and the Gaussian tail bound $\bar{\Phi}(x) \geq \frac{x}{1+x^2} \frac{e^{-x^2/2}}{\sqrt{2\pi}}$ gives $2^T \bar{\Phi}(\sqrt{T}(s_c - \varepsilon)) \geq cT^{-1/2} e^{T(\varepsilon s_c - \varepsilon^2/2)}$ for a constant $c > 0$, which grows exponentially for $0 < \varepsilon < 2s_c$, so the probability is summable. By the Borel-Cantelli lemma $M_T/\sqrt{T} \rightarrow s_c$ almost surely, i.e. $T^{-1} \max_Y (-E_{T,\omega}(Y)) \rightarrow s_c$; the minimum follows from the symmetry $E \mapsto -E$.

- (iii) Fix $s > 0$ (the case $s < 0$ follows from $E \mapsto -E$, and $s = 0$ is $\mathcal{Z}_0 = 2^T$). *Frozen upper bound.* $\mathcal{Z}_{s,T,\omega} \leq 2^T e^{s \max_Y (-E)}$, so by (ii), almost surely $\limsup_T T^{-1} \log \mathcal{Z}_{s,T,\omega} \leq a + ss_c$. *Annealed upper bound.* $\mathbb{E} \mathcal{Z}_{s,T,\omega} = 2^T e^{s^2 T/2}$, so by Jensen's inequality $\mathbb{E} \log \mathcal{Z}_{s,T,\omega} \leq T(a + s^2/2)$. To pass from the mean to the sample, note that $F(E) = \log \sum_Y e^{-sE(Y)}$ has gradient $-s\pi_s$ with π_s a probability vector, so F is $|s|$ -Lipschitz for the Euclidean norm of $E \in \mathbb{R}^{2^T}$, and $E = \sqrt{T}G$ with G a standard Gaussian vector; the Gaussian concentration inequality of Tsirelson, Ibragimov and Sudakov (Ref. [1], Theorem 5.6, applied to the $|s|\sqrt{T}$ -Lipschitz

function $G \mapsto F(\sqrt{T}G)$ gives

$$\Pr(|\log \mathcal{Z}_{s,T,\omega} - \mathbb{E} \log \mathcal{Z}_{s,T,\omega}| > \varepsilon T) \leq 2e^{-\varepsilon^2 T / (2s^2)}, \quad (\text{S5})$$

which is summable; hence almost surely $\limsup_T T^{-1} \log \mathcal{Z}_{s,T,\omega} \leq a + s^2/2$. Together, $\limsup \leq \min(a + s^2/2, a + ss_c)$; this is $\varphi(s)$ for $s \leq s_c$ but exceeds $\varphi(s) = ss_c$ for $s > s_c$, and the sharp value comes from the lower bound. *Lower bound by truncation.* Fix $0 < e < s_c$ and $\delta > 0$ with $e + \delta < s_c$, and let N_T be the number of records with $-E_{T,\omega}(Y) \in [Te, T(e + \delta)]$, a binomial variable with 2^T trials and success probability $\pi_T = \bar{\Phi}(\sqrt{T}e) - \bar{\Phi}(\sqrt{T}(e + \delta)) \geq c'T^{-1/2}e^{-T(e+\delta)^2/2}$ for large T and some $c' > 0$. Then $\mathbb{E}N_T = 2^T \pi_T \geq c'T^{-1/2}e^{T(a-(e+\delta)^2/2)}$ grows exponentially, and Chebyshev's inequality gives $\Pr(N_T < \frac{1}{2}\mathbb{E}N_T) \leq 4\text{Var } N_T / (\mathbb{E}N_T)^2 \leq 4/\mathbb{E}N_T$, which is summable. Almost surely, therefore, for all large T , $\mathcal{Z}_{s,T,\omega} \geq N_T e^{sTe} \geq \frac{1}{2}\mathbb{E}N_T e^{sTe}$, i.e. $\liminf_T T^{-1} \log \mathcal{Z}_{s,T,\omega} \geq a - (e + \delta)^2/2 + se$. Letting $\delta \downarrow 0$ along a sequence and taking the supremum over rational $e \in (0, s_c)$ (countably many events, each of full probability),

$$\begin{aligned} \liminf_T \frac{1}{T} \log \mathcal{Z}_{s,T,\omega} &\geq \sup_{0 \leq e \leq s_c} \left(a - \frac{e^2}{2} + se \right) \\ &= \begin{cases} a + s^2/2, & s \leq s_c, \\ a - s_c^2/2 + ss_c = ss_c, & s \geq s_c, \end{cases} \end{aligned}$$

because the concave function $e \mapsto a - e^2/2 + se$ is maximized at $e = s$ when $s \leq s_c$ and at the endpoint $e = s_c$ otherwise, and $a = s_c^2/2$. Upper and lower bounds agree, which proves Eq. (S2) for each fixed s on an event of probability one. *Uniformity.* Each $s \mapsto T^{-1} \log \mathcal{Z}_{s,T,\omega}$ is convex (a log-sum-exp), so convergence on the countable dense set $\mathbb{Q}$ to the finite convex function φ implies convergence for every real s , locally uniformly (Ref. [2], Theorem 10.8); the intersection of countably many full-probability events has probability one. The stated regularity of φ is read off from Eq. (S2): the two branches and their first derivatives (s and $s_c \text{sgn } s$) agree at $|s| = s_c$, the second derivatives do not.

(iv) By Eq. (S1), $\sum_Y P_{T,\omega}(Y)^{1-k} = \mathcal{Z}_{1-k,T,\omega} / \mathcal{Z}_{1,T,\omega}^{1-k}$, so $\phi_{T,\omega}(k) = T^{-1} \log \mathcal{Z}_{1-k,T,\omega} - (1-k)T^{-1} \log \mathcal{Z}_{1,T,\omega}$, and (iii) gives Eq. (S3) on the same full-probability event, simultaneously for all k ; Eq. (S4) is the identity $\tau(q) = -\psi(1-q)$ of Theorem 7 of the main text with $\varphi(1) = a + \frac{1}{2}$ and Eq. (S2) at $s = q \geq 0$. The regularity statements follow from those of φ ; in particular $\tau'(q) = a + \frac{1}{2} - q$ on $[0, q_c]$ and $\tau'(q) = a + \frac{1}{2} - s_c$ on $[q_c, \infty)$, continuous at q_c , and $\tau'' = -1$, respectively 0. The quadratic form of ψ on $[k_c, 1 + s_c]$ is $\varphi(1-k) - (1-k)\varphi(1) = a + \frac{1}{2}(1-k)^2 - (1-k)(a + \frac{1}{2}) = \frac{1}{2}k^2 + (a - \frac{1}{2})k$.

(v) ψ is finite, convex and differentiable on all of $\mathbb{R}$, hence essentially smooth (Ref. [3], Definition 2.3.5), and 0 is interior to its domain, so all hypotheses of Theorem 7 of the main text hold for the sequence $(P_{T,\omega})_T$, whose

finite-horizon log-MGFs are $\phi_{T,\omega}$ by (i). The Legendre transform of the quadratic branch $\frac{1}{2}k^2 + (a - \frac{1}{2})k$ is $\frac{1}{2}(h - h^*)^2$ with $h^* = a - \frac{1}{2}$, attained at $k = h - h^*$, which lies in $[k_c, 1 + s_c]$ exactly when $h \in [h_{\min}, h_{\max}]$ with $h_{\min} = k_c + h^* = a + \frac{1}{2} - s_c$ and $h_{\max} = 1 + s_c + h^* = a + \frac{1}{2} + s_c$. On the affine branches $\psi(k) = h_{\min}k + (\psi(k_c) - h_{\min}k_c)$ for $k < k_c$ and $\psi(k) = h_{\max}k + (\psi(1 + s_c) - h_{\max}(1 + s_c))$ for $k > 1 + s_c$ (the slopes are $\psi'(k_c)$ and $\psi'(1 + s_c)$ by continuity of ψ'), so $\sup_k [kh - \psi(k)] = +\infty$ for $h < h_{\min}$ and for $h > h_{\max}$. This gives I as stated and, by $f(\alpha) = \alpha - I(\alpha)$ (Theorem 7 of the main text), $f(\alpha) = \alpha - I(\alpha)$; $f(h_{\min}) = h_{\min} - \frac{1}{2}(h_{\min} - h^*)^2 = h_{\min} - \frac{1}{2}k_c^2$, and indeed $h_{\min} - \frac{1}{2}k_c^2 = (a + \frac{1}{2} - s_c) - \frac{1}{2}(1 - s_c)^2 = a - \frac{1}{2}s_c^2 = 0$. The concentration statement is the meaning of $f(h_{\min}) = 0$ together with Eq. (S4): for $q > q_c$, $\tau(q) = qh_{\min}$, so $Z_{q,\omega}(T) = e^{-Tqh_{\min} + o(T)}$ is carried by the records of minimal surprisal rate. $\square$

SVI. PROOF OF THE FINITE-POPULATION NORMALIZATION PROPERTY (PROPOSITION OF THE MAIN TEXT)

The main text defines the bootstrap estimator for the record SCGF and states its finite-population normalization property, deferring the proof here. For a fixed time-homogeneous environment with outcome kernel $K_\omega(x, dy)$, update $F_\omega(x, y)$ and additive increment $g_\omega(x, y)$ (in the record application, $g = -\log r$), the Feynman-Kac operator of the main text is

$$(\mathcal{Q}_{k,\omega}f)(x) = \int K_\omega(x, dy) e^{kg_\omega(x,y)} f(F_\omega(x, y)). \quad (\text{S6})$$

The estimator: initialize N normalized conditional states; at event t , independently for each particle i , sample Y_t^i from its Born predictor, update and normalize the state, and set $\ell_t^i = -k \log r_t^i$ (for a compound block, r_t^i is the product of its sequential conditional Born probabilities and $-\log r_t^i$ their sum). With resampling at every declared event, compute

$$m_t = \max_i \ell_t^i, \quad \log c_t = m_t + \log \left(\frac{1}{N} \sum_i e^{\ell_t^i - m_t} \right), \quad \bar{w}_t^i = \frac{e^{\ell_t^i - m_t}}{\sum_j e^{\ell_t^j - m_t}}$$

then resample with a conditionally unbiased scheme and reset weights. The estimates are

$$\hat{Z}_T(k) = \prod_{t=1}^T c_t, \quad \hat{\psi}_{L,\omega}(k) = \frac{1}{T} \sum_{t=1}^T \log c_t. \quad (\text{S7})$$

Under conditionally unbiased resampling, $\hat{Z}_T(k)$ in Eq. (S7) is unbiased for $Z_T(k) = \mathbb{E}_{P_\omega} e^{kA_T}$ at every finite N, T [4]; in general $\log \hat{Z}_T$ is biased; and if outcomes are drawn from an auxiliary proposal Q_t rather than the Born kernel K_t , the incremental weight must be

$$w_t = e^{kg(x,y)} \frac{K_t(x, dy)}{Q_t(x, dy)}. \quad (\text{S8})$$

Proof. Let $\gamma_t(f) = \mathbb{E}_{P_\omega}[f(X_t)e^{kA_t}]$ be the unnormalized Feynman–Kac measure, so that $Z_t(k) = \gamma_t(1)$ and $\gamma_t(f) = \gamma_{t-1}(\mathcal{Q}_{k,\omega}f)$ with the operator of Eq. (S6) (the increment $g = -\log r$ gives $e^{kg} = e^\ell$). Let $\eta_t^N = N^{-1} \sum_i \delta_{x_t^i}$ be the equally weighted particle measure after resampling at event t , $\hat{\eta}_t^N = \sum_i \bar{w}_t^i \delta_{x_t^i}$ the weighted one before resampling, and $\mathcal{F}_{t-1}$ the history up to the resampling at event $t-1$. Because particle i draws Y_t^i from its own Born predictor independently of the others, $\mathbb{E}[e^{\ell_i} f(x_t^i) \mid \mathcal{F}_{t-1}] = (\mathcal{Q}_{k,\omega}f)(x_{t-1}^i)$; averaging over i and using $c_t \hat{\eta}_t^N(f) = N^{-1} \sum_i e^{\ell_i} f(x_t^i)$ gives $\mathbb{E}[c_t \hat{\eta}_t^N(f) \mid \mathcal{F}_{t-1}] = \eta_{t-1}^N(\mathcal{Q}_{k,\omega}f)$. Conditionally unbiased resampling means $\mathbb{E}[\eta_t^N(f) \mid \hat{\eta}_t^N] = \hat{\eta}_t^N(f)$. Multiplying by the $\mathcal{F}_{t-1}$ -measurable $\hat{Z}_{t-1} = \prod_{s < t} c_s$ and combining the two conditional expectations,

$$\mathbb{E}[\hat{Z}_t \eta_t^N(f)] = \mathbb{E}[\hat{Z}_{t-1} \eta_{t-1}^N(\mathcal{Q}_{k,\omega}f)].$$

By induction from $\hat{Z}_0 = 1$ and $\eta_0^N = \delta_{x_0} = \gamma_0$ this gives $\mathbb{E}[\hat{Z}_t \eta_t^N(f)] = \gamma_0(\mathcal{Q}_{k,\omega}^t f) = \gamma_t(f)$ for every bounded f , and $f = 1$ yields $\mathbb{E}\hat{Z}_T = Z_T$ (Ref. [4], Theorem 7.4.2). Jensen’s inequality gives $\mathbb{E} \log \hat{Z}_T \leq \log Z_T$, with equality only if $\hat{Z}_T$ is almost surely constant, so the logarithm is biased in general. Equation (S8) is the Radon–Nikodym change-of-measure identity: with outcomes drawn from Q_t , the conditional expectation of $w_t f(x')$ is $\int Q_t(x, dy) e^{kg(x,y)} \frac{K_t(x, dy)}{Q_t(x, dy)} f(F(x, y)) = (\mathcal{Q}_{k,\omega}f)(x)$, and without the ratio the proposal expectation is not the target Feynman–Kac integral. $\square$

SVII. THE CLIFFORD RECORD-COUNT RUNS

This section documents the runs behind the main-text Sec. IV.F (the disorder SCGF of the Clifford record count): the exact evolution and its calibration, the trajectory simulator and its validation, the production ladder, the estimator definitions, and the reproduction commands.

A. The exact 3^L algebra evolution and the deposit calibration

The annealed quantities are computed exactly by evolving the two-replica state on the per-site algebra $\{e_0, e_+, e_-\}^{\otimes L}$ (the 3^L product basis): the gate-bond twirl is the exact factorised projection $\alpha I + \beta S$ with $\alpha = (4t_i t_j - s_i s_j)/60$, $\beta = (4s_i s_j - t_i t_j)/60$ ($d = 2$; per-site (t, s) : e_0 : (2, 2), e_+ : (1, 1), e_- : (1, -1)), the measurement layer is diagonal ($e_0 \mapsto e_0$, $e_\pm \mapsto (1-p)e_\pm$: the measure/no-measure choice is a single event shared by both replicas—the naive $(1-p)^2$ damping was caught and corrected by the deposit calibration below), and the initial state $|0 \cdots 0\rangle\langle 0 \cdots 0|^{\otimes 2}$ maps, after the (inert) first measurement layer and the first gate layer, to

$\otimes_{\text{even bonds}} [(e_0 + e_+)^{\otimes 2} + e_-^{\otimes 2}]/10$ (trace 1 exactly). The observables are linear functionals: $Z_2 = \sum_a c_a \prod_k t_{a_k}$ and $P_{2,A} = \sum_a c_a \prod_k t_{a_k} \prod_{k \in A} \varepsilon_{a_k}$ with $\varepsilon = (1, 1, -1)$, the last being the exact algebraic form of the replica swap S_A .

The calibration (Table S10) fixes the deposited chain’s time convention: the deposited chain runs at $t = L/2$ periods ($t = 4$ at $L = 8$, $t = 6$ at $L = 12$); the “ $t = 4L$ ” reading of this chain is excluded by the calibration (at $t = 4L = 32$ one instead gets $Z_2 = 4.57 \times 10^{-18}$, twenty-nine orders away). The deposited $L = 12$ quenched entropy 2.394(5) is not reproduced at either convention (our chains: 2.497(7) at $t = 4$, 2.632(5) at $t = 6$) while both deposited annealed values are matched at $t = 6$; it is quoted in the main-text Appendix B as deposited and flagged here as a provenance exception on which nothing in the main text depends. The trajectory ensemble is an independent implementation (vectorised phase-free stabilizer tableau, uint64 x/z words, the 720 two-qubit Clifford actions as lookup tables, M–A–M–B period order, new PCG64 seed contract `seed0 = 20260529 + cell`) and reproduces both the exact Z_2 (relative deviation 0.73% and 0.59% at $L = 8, 12$) and the deposited quenched chain at $L = 8$.

B. Exact anchors and the production ladder

The $\beta = 1$ trajectory estimates degrade with the ESS collapse (e.g. -0.08037 at $L = 8$ against the exact -0.07798 , the bias growing with L), which is why Table S12 reports the exact anchors in the $\Lambda_L(1)$ column; the trajectory columns are used only for $\beta \leq 0.75$, where the moment estimates are well conditioned. The $\Lambda_L(2)$ trajectory estimates ($L = 8, 12, 16$ at $p = 0.16$: -0.14486 , -0.15891 , -0.17019 ; at $p = 0.22$: -0.20652 , -0.22329 , -0.23247) underlie the ESS law of the main text; $\Lambda_L(2)$ is exactly the $n = 3$ annealed moment $(2Lt)^{-1} \ln \bar{Z}_3(t)$ of the main-text Eq. (dyadic moments) and is in principle computable exactly by the three-replica operator.

C. Estimators, error model, and reproduction

The trajectory estimator of $\Lambda_L(\beta)$ at fixed (L, p, t) is the logarithm of the sample mean of $2^{-\beta N_b}$ over B independent disorder draws b (each draw contributing one record under its own Born measure; $N_b = N_{\text{rand}}(\omega_b)$ is ω_b -measurable, so one record per draw suffices), with bootstrap standard errors over 200 trajectory resamples; the tilted densities are the weighted means $\tilde{m}(\beta) = \sum_b 2^{-\beta N_b} N_b / \sum_b 2^{-\beta N_b} / (2Lt)$; the ESS is the within-sample ratio $(\sum_b w_b)^2 / \sum_b w_b^2$. The raw arrays (X_R and $S_{L/2}$ per trajectory, all eight production cells) are deposited as `scgf_born_raw_L{8,12,16,24}_p{16,22}.npz`.

Reproduction (from the `research/scripts/` directory):

command	purpose
<code>python3 mipt_scgf_exact.py</code>	the 3^L evolution: calibration, anchors, and subexponential (Table S10, S11)
<code>python3 mipt_born_scgf.py</code>	the trajectory suite: validation, production ladder, ESS, tilted densities (Table S12)

Dependencies: `numpy` (PCG64). Wall time: the full suite is minutes on one core except the $L = 16$ exact evolution (3^{16} configurations, hours). The files of this round are hashed in `certificate_sha256_v8.txt` (round files) and `certificate_sha256_v3_scgf.txt` (the parallel-line ledger of the record-SCGF suite itself).

D. The exact-closure runs: $\bar{Z}_3$, the Clifford design, and the $n = 5$ $L = 8$ block

a. The shared library. The shared module `gap_utils.py` (the Weingarten channel $W_{p,n}$ of Eq. (Wpn), the group utilities, and the dense bond-label operators `bond_ops`), on which the earlier script suites depend, is included in the deposit, exactly following Eq. (Wpn) (inverse of the Gram $D^{c(\sigma^{-1}\tau)}$ for $D \geq n$, Moore–Penrose pseudo-inverse otherwise), both in `scripts/v22-exactZ3-n5L8/` and as copies alongside the suites that use it. It is validated by re-running the deposited batteries unmodified: `n3_annealed_validate_v1.py` V1–V6 all PASS (the Sec. 9 benchmarks to 0.0×10^0 at six decimals, the Gram similarity 1.7×10^{-16} , the Kaufman closed form to 4×10^{-11} , the momentum-block union 1.3×10^{-16} , the colour sectors to 2×10^{-6} , the product rows exact) and `n45_annealed_validate_v1.py` V1–V8 ALL OK (including the $n = 5$ $p = 1$ bond eigenvalue $1/14$ through the pseudo-inverse channel and the pinv property $G \text{Wg} G = G$ to 3×10^{-15}).

b. The conjugation-design test. `v22_clifford_3design_v1.py` enumerates the two-qubit Clifford group by BFS closure from $\{H \otimes I, I \otimes H, S \otimes I, I \otimes S, \text{CNOT}_{01}, \text{CNOT}_{10}\}$: 92,160 unitaries, the phase extension $\{e^{ik\pi/4}\} \times \text{Cl}_2$; phase classes are deduplicated by the injective phase-invariant key $U \mapsto U \otimes \bar{U}$ (11,520 classes), and the census of the unsigned-Pauli actions gives exactly the 720 symplectic classes with 16 elements each. The test averages $C^{\otimes n} X C^{\dagger \otimes n}$ over the classes and compares with the Haar twirl $\sum_{\pi} U_{\pi} y_{\pi}$, $\text{Gy} = t$, $\text{G}[\pi, \rho] = (d^2)^{c(\pi^{-1}\rho)}$, on six random X at each n : $n = 2$ deviates 2.1×10^{-14} , $n = 3$ deviates 1.6×10^{-14} (the identification of Remark rem:cliff3design), $n = 4$ deviates 0.578 (the negative control).

c. The exact $\bar{Z}_3$ cells and ladder. `v22_z3_exact_v1.py` implements the boundary-vector formula of Prop. prop:z3closure on the compressed bond space with the `ring_gemm_v2` kernels. Validation: the $n = 2$ anchors of `scgf_exact.json` to 2.6×10^{-15} , the A -tail $2.4509 \rightarrow 2.4512$, the layer-parity invariance to 0.0. Production cells ($t = 4L$): Table tab:z3exact of the main text. The ladder: $\lambda_1^{(3)}$ by power iteration (vali-

dated against the dense $L = 8$ spectrum to 1.9×10^{-10}), with amplitudes $A_3(L)$ constant to six digits for $t \geq L$ and subexponential in L (Table S10, S11).

d. The $\beta = 2$ cross-check. `v22_traj_beta2_v1.py` runs the deposited tableau simulator at $t = L/2$ with $B = 4 \times 10^6 / 2.4 \times 10^6 / 1.6 \times 10^6$ trajectories at $L = 8/12/16$ (fresh PCG64 seeds, recorded): $\hat{\mathbb{E}}[2^{-2N_{\text{rand}}}]$ against the exact $\bar{Z}_3(t)$ and $\hat{\mathbb{E}}[2^{-N_{\text{rand}}}]$ against $\bar{Z}_2(t)$, with bootstrap standard errors and the empirical tilted ESS. The results are quoted in the main text (Prop. prop:z3closure).

e. The marginal $q = 4$ log-correction fits. `v22_n4_eps_topup_v1.py` extends the $L = 8$ λ_{ϵ} scan to $p = 0.39, 0.40, 0.41$ (the deposited $p = 0.38$ value reproduced to 5.7×10^{-16} first; the $p = 0.38$ cell supplies the $n = 4$ $L = 8$ rung row of the main-text size-ladder table). `v22_q4_logcorr_v1.py` then evaluates R_L , the slopes and the gaps at the common reference $p^* = 0.383$, and fits the marginal forms: $R_L = 1/4 - 0.90/\ln L + 0.46/\ln^2 L$ (residual $\leq 8 \times 10^{-4}$, against the corrected $q = 4$ target $x_{\sigma}/x_{\epsilon} = 1/4$); $dX_L/dp = C L^{3/2}(1 + 0.45/\ln L)$; the $n = 3$ control converging cleanly to $1/6$. The crossing table is recomputed with the top-up (p_c crossings $0.3582/0.3790/0.3823$; $1/\nu_{\text{eff}} 1.00 \rightarrow 1.38 \rightarrow 1.57$).

f. The tilted-variance profile family. `v22_tiltvar_uniform_v1.py` measures $\text{Var}_s(\log X)$ on $s \in [0, 1.5]$ (61 points, tilted empirically, 200 bootstrap resamples): Part A, exactly enumerated Haar $p = 1$ collision partition functions ($k^* = -1$) over the grid $L = 3-6$, $T = 2L, 4L, 8L$, 6000 circuits per cell; Part B, fresh Clifford disorder trajectories at $t = L/2$ (the same well-conditioned cells as the $\beta = 2$ cross-check). Metrics per cell: the per-period variance rate $v(0) = \text{Var}_0/T$ (T-independent within each L to 3%), the sub-Gaussian profile ratio $\rho = 2 \int_0^1 (1-s)v(s)ds/v(0)$, the initial decay rate κ , and the interpolation identity as a per-cell validation ($\leq 2.5 \times 10^{-6}$). Part A: $\rho = 1.02 \pm 0.03$ across the whole grid (a flat-to-mildly-rising profile, (L, T) -uniform in shape); Part B (Clifford disorder): $\rho = 0.872/0.878/0.888$ at $L = 8/12/16$, $p = 0.16$ ($0.890/0.894/0.897$ at $p = 0.22$): the sub-Gaussian decaying profile anticipated in the main-text Discussion, confirmed on well-conditioned data and mildly de-sub-Gaussianizing with L . The raw production $t = 4L$ arrays are not used for Part B: their tilted ESS collapses to 1, the same estimator difficulty as the $\beta = 2$ moment (the within-sample profiles are reported in the JSON for completeness).

g. The symmetry-reduced $n = 5$ $L = 8$ block route. `v22_n5_L8_block_v1.py` assembles the momentum-zero colour-trivial block of $\text{C}_{\text{comp}}^{(5)}$ at $L = 8$ exactly: the orbit basis of the symmetry group $(S_5 \times S_5) \rtimes \mathbb{Z}_4$ on $(S_5)^4$ (canonical forms = the minimum over conjugators and shifts of the encoded lkey tuple; the orbit table is p -independent and memmapped), the block elements $B[\alpha, \beta] = \sqrt{n_{\alpha}/n_{\beta}} \sum_{\tau \in O_{\beta}} M[\text{rep}_{\alpha}, \tau]$ by a chunked pass (segment sums over orbit-sorted chunks), and

$C_{\text{block}} = B_{M_1} B_{M_2}$. Validation: the deposited $n = 5$ $L = 4$ `triv3` rows reproduced to 1.6×10^{-14} ($p = 0.44$ and 0.32), and the $L = 6$ blocks against fresh unrestricted colour-restricted Arnoldi solves. The $L = 8$ rung—the third rung of the size ladder—runs the block at the locator grid; the completed rung (5 grid points) is deposited as `v22_n5_L8_rung.json` and hashed in `certificate_sha256_v10.txt`.

Reproduction: run the scripts in this order (each writes its JSON to `results/v22-exactZ3-n5L8/` and its log to `logs/v22-exactZ3-n5L8/`):
`v22_clifford_3design_v1.py`,
`v22_z3_exact_v1.py` all, `v22_traj_beta2_v1.py`,
`v22_n4_eps_topup_v1.py`, `v22_q4_logcorr_v1.py`,

`v22_tiltvar_uniform_v1.py`,
`v22_n5_L8_block_v1.py` validate,
`v22_n5_L8_block_v1.py` ids,
`v22_n5_L8_block_v1.py` run `--pgrid`
0.44,0.46,0.47,0.48,0.50. Dependencies: `numpy`,
`scipy` (dense eigensolvers); the wall time is minutes
per script except the $L = 8$ block assembly (hours;
the canonical-id pass is p -independent and done once).
The $L = 6$ transfer application runs in 5.4s (float64)
or 2.9s (float32) per matrix–vector product on two
cores, with < 500 MB peak memory. The exact-closure
files are hashed in `certificate_sha256_v9.txt`; the
 $L = 8$ rung result `v22_n5_L8_rung.json` (with the
manuscript and supplement of record) is hashed in
`certificate_sha256_v10.txt`.

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- [1] S. Boucheron, G. Lugosi, and P. Massart, *Concentration Inequalities: A Nonasymptotic Theory of Independence* (Oxford University Press, Oxford, 2013).
 - [2] R. T. Rockafellar, *Convex Analysis* (Princeton University Press, Princeton, 1970).
 - [3] A. Dembo and O. Zeitouni, *Large Deviations Techniques*

- and Applications*, 2nd ed. (Springer, New York, 1998).
- [4] P. Del Moral, *Feynman–Kac Formulae: Genealogical and Interacting Particle Systems with Applications* (Springer, New York, 2004).

TABLE S8. Certificate files and SHA-256 hashes.

file	SHA-256
n3_mult_cert_nb2.json	5600c82b82c7875210b9ab7803d0a5c8f6934d57f6afbc8f016679d6435ea56a
n3_mult_cert_nb3.json	135635657992709144b16324a470977ed394a02bc5b21c11bc2da130294807de
n3_mult_cert_nb4.json	2b618ab68941a68d53d6a0b678411f65053308c8eae8373a083aa55d155ad1dd
n3_mult_cert_nb2_polys.json	26fcc456ce8341f10d34c8c6abca08d04de9e0241984f7ccd43057d38e4a542a
n3_mult_cert_nb3_polys.json	016233445a4e8139c07dce862e1be358f8fa094df277517a942b14381b2eb837
n3_mult_cert_nb4_polys.json	9d944ec1dcacfb1a26216cae008fec698573cfa11de95935c084025abece505c
n3_mult_cert_d3_nb2.json	20cc93dfe011ab6fc20925a5310bdc3974f1aa5ab6853603c0e00e750cf89b30
n3_mult_cert_d3_nb3.json	7fcd62c254fff32a078dca08e268e16c254478918c6aa6c075d40228359bfe9
n3_mult_cert_d3_nb4.json	f258be716d31e4a1d48472bfcdb364852eaeefcb8b0629733e82c4d84e7d0be78
n3_mult_cert_d3_nb2_polys.json	161e1ef5fc91debe27f598a6aee68199dbd35419fdc8fd4217204a5c7505973a
n3_mult_cert_d3_nb3_polys.json	7e23016ad43897e250fc6e089b02a84fc0eb36814d06184a8ed0c9f6ecc625d0
n3_mult_cert_d3_nb4_polys.json	1295e4deb729ce426eccad936240c71f7272a096c24344ccacbc6b81fd90c7
n3_exceptional_exact_nb3.json	55effc4a80ba193af63f449c6599bbf590e3bb0286bd015a79650b635c9996fa
n3_exceptional_exact_nb4.json	32442fe7d0dd851619d9a8b4c349c5595abb37df7c020b4594419f5657d66f22
n3_exceptional_exact_d3_nb2.json	11ce1f740cbe4d5907461c4bb730d4fd5f7e67b1d4a0a496d1c52738d119e9ee
n3_exceptional_exact_d3_nb3.json	785850b46a07fbde476eda6f26058fcee7e56c7ac34745a5f9e2488b2fe7390e
n3_exceptional_exact_d3_nb4.json	e53211284b01a69f0f693296bff02d82fd5b4e42489f39012880100ab3bb8d1b
n3_L12_kernel_verify_k3.json	37c2581e753c771b51c84166f0decad50f9262028243d699ff7643a42b695666
n3_L12_moving_kernel_2_Ind(triv.sgn)_minus.json	451e548b6b90b150419091f5e829d0e10a7ef9cee5f1abd56485c2386c9d98a1
n3_L12_moving_kernel_4_Ind(sgn.std)_minus.json	e685aa5e8e927203094af86049bbb093fac6f3200dd63a30046b824d2664555b
n3_L12_moving_kernel_4_Ind(triv.std)_minus.json	9dd9a9d3cece96412693eefc9086858a03a2192402df6a442d7dd0a1a7f5321a
n3_L12_moving_kernel_4_std.std_i_minus.json	96c91b35b2fcb2905cfd0083d1217d12eba8900f09d979f1707c181943dbf55f
n3_L12_moving_kernel_2_Ind(triv.sgn)_minus_d3.json	51cf134d4969294c65ffdaa941e2b6dc7769ef473060218a7f6ca783d9842280
n3_L12_moving_kernel_4_Ind(sgn.std)_minus_d3.json	6eeb8cb6a8cb4f9a5734180e91eb0f720ee80a420a4ea9cdeb9e1b1f3c79abb7d
n3_L12_moving_kernel_4_Ind(triv.std)_minus_d3.json	32bc17daddb1981d9769e718825f6a92e7dae6c6c1963e71ffcac05ad0720e3d
n3_L12_moving_kernel_4_std.std_i_minus_d3.json	afe7d826252cc60b536647ad15fc9cc059494540cdbe59686a374937168527a8
n3_L12_kernel_1_sgn.sgn_i+_k3_q1048261_rational.json	0f49fed2443e23c4f446d46cdf30befa901ba16561561bc826d25d5f127e7e7b
n3_L12_kernel_1_sgn.sgn_i-_k3_q1048261_rational.json	4acd6bc8d9bc19682a595e82403dc175f9912d86cd6cecc5b15334dfecfc981a
n3_L12_kernel_1_triv.triv_i-_k3_q1048261_rational.json	1d2aa2f1901995a12c92e9acf19d0f71a87617f4a73dd7128ea7a09b35fec639
n3_L12_kernel_2_Ind(triv.sgn)_k3_q1048261_rational.json	9d02b012b4f1d9b631ec1bb305be7f7fea6c6c568a8ac12b295ba4926e939285
n3_L12_kernel_4_Ind(sgn.std)_k3_q1048261_rational.json	5652eebdda79b9a665da7ba07e77385dadb55a88df8f272404a4065da18bc530
n3_L12_kernel_4_Ind(triv.std)_k3_q1048261_rational.json	82e14c6e6396ee75589fd265d422143f39cfcc3597d463bc0026d0f8ac1fa
n3_L12_kernel_4_std.std_i+_k3_q1048261_rational.json	33be5db62064dbdc2088fff795f25b07ef3151060ea55e3d4f23a4f5fcb6b2e3
n3_L12_kernel_4_std.std_i-_k3_q1048261_rational.json	b5edc70bc1eb3f3d47c6b110b797d7ef84435f487c272fb9044e0ff4db9d4a45
n3_L12_final.json	0aa3f5a6e678b8bc80b0cba4d0a84a170a54cbdb810c9d7e666a42068c795fd75
n3_L12_final_d3.json	09dcd3fc785da31010975c85b7dcec2bafef6fd8cf62230882dd0ad06dae4aff
n3_sharpness_L10.log	b0969da19857824e0637ee1e1a634e45039d793019161eee1333f71397d0d81f
n3_sharpness_L12.log	f3c2f28467d4acf15ca952587bfbfb90ebf527b4d5e39e8299002fabd7ac0930
n3_sharpness_L12_k3_central.log	70664316b052b71c69e6ef4ac620e8f6309c65baea54e1232d68276ace414e8
n3_sharpness_L12_k3_d3.log	165d9079fb4a065396d0a49d9666143be90a6a36542c3230c0b95f15888868793
n3_sharpness_L12_k3_p1-2.log	c28642ef7b473425a9bdaee11f53440890a083c15e85a10e9ed8d91d17ed3f11
n3_sharpness_L12_k3_p1-3.log	568ca64a32e15eb8007b2971fb4740eab400c2ec8d73776d25fb053a7bc1d8bf
n3_sharpness_L12_k3_p3-5.log	129d75f4cf9e8e9a9a49ff57ee704d9850f0fec87842a087bc1693bcb6d1b85
n3_sharpness_L12_d3_full.log	36060cf646724100fd3bf8af19b28c7ba8a8dcb32057ec3bbc4d7332aafa3cf6
n3_sharpness_L12_p0.log	53dfefbc3ccbf5a3dd0707b942640464528b497140e1bea339bcfd2c0dea1889
n3_sharpness_L12_p0_q1048573.log	1c7b9af12a9e6a3f9c4cfcb9070f73830c8dde1425dfff2e0377580f94e6bf6e
n3_sharpness_L12_d3_px.log	17fab566985c51be05fe068853813704e9e25fbb758ed95180cb96dd92e6aa0
n3_sharpness_L12_d3_px_q1048573.log	34532374c00dfa678be17c2ffe0c8cf9ae88ffd06b3585cbf307822e0efa865e
n3_sharpness_L10_d3.log	a1663286d77125fa8cf673084dce411df512aa9b3dd38e7fe9d4d976dbfe771d
n3_sharpness_L10_d4.log	af8c189121d9d15692f0add848aea65d63d96d27b52920e4ccca1c8902c9e2b7
n3_sharpness_L10_d5.log	a5999f67c25a9165dd946a9e5949fa185d17924e79100918e63a8530f7664234
n3_sharpness_L10_d6.log	10292421bfc5316cd1fd28ca5f22d125f2fcc340773f5f7001bf03fd8e75a68b
n3_sharpness_L10_d5_p1-3.log	cf435e2612e156df6600114189675f1853528bcbf4edf2b2281f95644f8b143e
n3_sharpness_L4-10_d7_p1-3.log	8b76837e6edfe453030ec5633a841661efed12a696f1d6fc36529873049096d6
n3_sharpness_L4-8_d456_p1-3.log	9f344ea8e8d7df8501d722fb23ecf0f3d02a2b637d17bbc2990d7850e084d18a

TABLE S9. Scripts and SHA-256 hashes.

script	SHA-256
gap_utils.py	784e2aa0e2275c8fa9123073a1ac267b6ca71279231801a484b7cbf410e1a3d1
n3_mult_certificates.py	ed2abc22e9dbcc091372e1845b8a9ca2fcc0cd271ec4f9c01040a37f77b9b561
n3_mult_certificates_d.py	81e8b91b0de3df0140edbc21848ccd7d42923c0b76331cc182b7e5b7f51825c
n3_exceptional_exact.py	7b82757b018b9ac74ea963f66091eda3034c363aef842a0e74a1fd1fceb5cadd
n3_exceptional_exact_d.py	1753756feddbce19e4f630e933d0fcbe169e3a2c46776be41ac73a0a5677a791
n3_simple_zero_check.py	30fc0dfa4795e5e637b252516f48b95c0f5f9e7dbcdabb4e703ffab2fd9d9cb35
n3_p0_exact_rank.py	adedb74bf3b3956cd4d2b77392950475a2c80da7bb449f776b3bf69c0e8ba582
n3_closed_form_proof.py	da8cc292493b9d7bc030b13a31a81b8289fd5e940beca19900b05e4abb3eb5d2
n3_formula_exact.py	2accd8a1e5360fcba4982138b4973ed2f4848cb1cb7b270f91d85dacc144dfe3
n3_sharpness_L12.py	3dffec312836e13489c48889d3209a862c207314f2fbc3ff6f63cebd8cf350d3
n3_L12_structure_check.py	bd5064ad249d9589f5200ec9e4c5203392176391f46014f3aec46a15d8e7d312
n3_L12_kernel.py	dd962b398a3888f1c6b1c26368c91a9af59c63a5d9ebc3a302f4f0e53ae581e5
n3_L12_kernel_verify.py	548dd40a29b15c7d0fd8188f1e750aeb82fb210006054bcfed40f6fa949cad0f
n3_L12_moving_kernel.py	07b13b7a3eac81d647290a15c86fd122bcacdc2db7ba40bea868f193550800be
n3_L12_final.py	613584dd468066855a842c6ddcf42535bb8d428bfd9de18659422d0f742e9087
n3_kernel_recheck.py	30062ba34504253b2d770d5bfb795d38371ad1aeab1bafc92fd8a467c142079

TABLE S10. Calibration of the exact evolution and of the trajectory simulator against the deposited tilt chain (script `mipt_scgf_exact.py`, run logs `mipt_scgf_exact.log`, `mipt_born_scgf.log`). The deposited exact values are reproduced to their rounding at both sizes.

quantity	$L = 8, t = 4$	deposit	$L = 12, t = 6$	deposit
Z_2 (exact evolution)	1.479639×10^{-2}	1.4796×10^{-2}	3.81731×10^{-5}	—
$\tilde{S}_2$ (exact)	2.1547	2.155	3.1281	3.128
$\mathbb{E}[2^{-N_{\text{rand}}}]$ (4×10^4 traj.)	1.4689×10^{-2}	1.4942×10^{-2}	3.8399×10^{-5}	—
$\mathbb{E}S_{L/2}$ (traj.)	2.015(4)	2.019(4)	2.632(5)	2.394(5)
ESS (tilt)	6339	5935	478	454

TABLE S11. Exact $\beta = 1$ anchors of Λ_L (`scgf_exact.json`, `scgf_exact_addendum.json`): finite- t values at $t = \tau L$ from the 3^L evolution ($L \leq 16$), the asymptotic $\log \lambda_1(L)/(2L)$, and the tail amplitude $A(L) = \tilde{Z}_2(t)\lambda_1^{-t}$ (constant to six digits for $t \gtrsim 3L$; `z2_tail` in `scgf_exact.json`).

L	p	t	$2Lt \Lambda_L(1)$ fin. t	$\log \lambda_1/(2L)$	$A(L)$
8	0.16	32	-0.077984	-0.079735	2.4512
12	0.16	48	-0.079362	-0.080615	4.2354
16	0.16	64	-0.079901	-0.080867	7.2288
24	0.16	96	-0.080325 ^a	-0.080989	$\approx 21.4^b$
8	0.22	32	-0.106382	-0.108701	3.2786
12	0.22	48	-0.108278	-0.109947	6.8326

^aAmplitude-corrected λ_1 anchor, $\log \lambda_1/(2L) + \log A/(2Lt)$.

^bExtrapolated from $\log A(L) \approx 0.135 L$
($\log A = 0.896, 1.443, 1.979$ at $L = 8, 12, 16$).

TABLE S12. The production ladder (`scgf_born.json`; $t = 4L$; $\bar{x}_L$ and $\text{Var}/(2Lt)$ in bits per site and bits² per site; Λ_L values in nats per site; bootstrap s.e. (200 resamples) on $\bar{x}_L$ at the 10^{-3} level). g_L uses the exact finite- t anchors of Table S11 ($L \leq 16$) and the amplitude-corrected value at $L = 24$. The ESS column is the within-sample collision-tilt effective sample size at $\beta = 1$.

L	p	B	$\bar{x}_L$	$\text{Var}/(2Lt)$	$\Lambda_L(0.25)$	$\Lambda_L(0.5)$	$\Lambda_L(1)$	g_L	$\text{ESS}_{\beta=1}$
8	0.16	40000	0.14479	0.10547	-0.02357	-0.04440	-0.07798	0.02238	11.7
12	0.16	40000	0.14703	0.10441	-0.02392	-0.04495	-0.07936	0.02255	3.9
16	0.16	30000	0.14803	0.10581	-0.02412	-0.04586	-0.07990	0.02270	1.4
24	0.16	15000	0.14890	0.10481	-0.02428	-0.04689	-0.08033 ^a	0.02288	1.0
8	0.22	40000	0.19280	0.12532	-0.03158	-0.05988	-0.10638	0.02726	18.2
12	0.22	40000	0.19512	0.12415	-0.03203	-0.06110	-0.10828	0.02697	2.7
16	0.22	30000	0.19614	0.12409	-0.03219	-0.06165	—	—	1.1
24	0.22	15000	0.19706	0.12365	-0.03225	-0.06274	—	—	1.0